[SYNTHETIC PIPELINE SANITY TEST — NOT REAL LITERATURE]

Interleukin-6 Triggers STAT3 Phosphorylation and Negative Feedback Regulation by SOCS3

Authors: Synthetic Benchmark Generator
Provenance: Synthetic Pipeline Sanity Test (Procedural Fixture)

Page 1 — Abstract

Interleukin-6 (IL-6) is a pleiotropic cytokine regulating inflammatory responses and hematopoiesis. Excessive IL-6 signaling contributes to autoimmune pathology.

Page 2 — Introduction

Binding of IL-6 to its receptor complex induces gp130 dimerization. This receptor engagement rapidly activates JAK2 tyrosine kinase at the inner membrane leaflet.

Page 3 — Methods

Activated JAK2 directly phosphorylates STAT3 at Tyr705. Phosphorylated STAT3 undergoes homodimerization and translocates into the nucleus.

Page 4 — Results

Quantitative RT-PCR and immunoblotting revealed that activated STAT3 strongly upregulates SOCS3 gene expression within 60 minutes of stimulation.

Page 5 — Discussion

Elevated SOCS3 protein binds JAK2 to downregulate kinase activity, establishing a negative feedback loop. Thus, IL-6 triggers SOCS3 expression through STAT3 phosphorylation.

Page 6 — Conclusion

Our study confirms that SOCS3 prevents excessive inflammation by attenuating continuous STAT3 signaling. Unstimulated controls did not induce SOCS3 expression.